PawMatch Post Development Testing

These tests were carried out on the current PawMatch project after implementation. Backend and database tests used a fresh temporary SQLite database, so the real project database was not changed. The application was also started and visually checked at desktop and mobile widths.

Outcome

All 16 completed tests passed. This provides functional, boundary, erroneous, security, database, algorithm and basic interface evidence. Stakeholder usability testing is listed separately because it must be completed by real users rather than simulated.

Completed tests

ID	Test and test data	Expected result	Actual result	Source	Result
T01	**Python modules compile** Compile all 16 Python files.	Every file compiles without a syntax error.	All 16 Python files compiled.	Q6	Pass
T02	**Registration validation** Validate registration with age 18, 17 and 121; valid/invalid email; short, overlong and mismatched passwords.	Valid boundary is accepted; invalid data gives field errors.	Valid age 18 was accepted and normalised. Age 17, age 121, invalid email, short/overlong password and mismatched confirmation were rejected.	Q9 Q12	Pass
T03	**Account creation and login** Create a user, repeat the email, then log in with correct and incorrect passwords.	One account is created; duplicate and incorrect login are rejected.	One user was created; duplicate email was rejected; correct credentials authenticated; wrong password returned no user.	Q9 Q12	Pass
T04	**Password hashing** Hash the same password twice and verify correct/wrong passwords.	Hashes differ, contain no plain password and verify correctly.	Two hashes for the same password were different; neither contained the password; verification accepted only the correct password.	Q12	Pass
T05	**Administrator role** Create an administrator and authenticate using the username.	The authenticated account has the admin role.	Administrator account authenticated by username and returned role 'admin'.	Q5 Q12	Pass
T06	**Animal record validation** Validate a complete animal, then blank name, age -1, housing 4, 25 hours alone, short description and invalid URL.	Complete data is accepted; every invalid field is rejected.	Complete animal data was accepted. Blank name, age -1, housing 4, 25 hours alone, short description and invalid URL were rejected.	Q2 Q15	Pass
T07	**Animal create edit delete** Insert Nova, change activity Moderate to High, read the record, then delete it.	Create, read, update and delete all persist correctly.	Nova was inserted, changed from Moderate to High activity, read back correctly, deleted and then no longer found.	Q5 Q15	Pass
T08	**Animal search and filters** Search 'Atl'; filter Cat + Small; search an unknown value.	Atlas only; Bramble only; then an empty result.	Partial search 'Atl' returned only Atlas; Cat + Small returned only Bramble; an unknown term returned an empty list.	E1 E2 E3 Q8	Pass

Completed tests continued

ID	Test and test data	Expected result	Actual result	Source	Result
T09	**Questionnaire validation** Submit valid questionnaire boundaries, then housing 0, activity 4, 25 hours and unsupported species/size.	Valid answers pass and invalid answers are rejected.	Valid boundary values were accepted. Housing 0, activity 4, 25 hours alone and unsupported species/size were rejected.	Q1 Q10 S5	Pass
T10	**Hand-calculated strong match** Calculate Atlas using a fully compatible adopter profile.	72/72 points = 100%, Strong, preference score 15.	Atlas received 100.0% (72/72 points), Strong category and preference score 15.	Q10 Q15 S7	Pass
T11	**Hand-calculated unsuitable match** Calculate Atlas using conflicting space, garden, time and activity answers.	15/72 points = 20.8%, Low, with four mismatch explanations.	Conflicting answers produced 20.8% (15/72 points), Low category, no preference bonus and four unmet-requirement explanations.	Q3 Q10 S9	Pass
T12	**Saved response and recommendation ranking** Save the questionnaire and rank Atlas, Bramble and reserved Clover.	Atlas then Bramble; reserved Clover is excluded.	Response 1 was saved. Recommendation order was ['Atlas', 'Bramble']; reserved Clover was excluded.	Q5 Q10 Q14 S7 S9	Pass
T13	**User data isolation** Save different questionnaire responses for two users.	Each user retrieves only their own latest answers.	Two users retained separate latest questionnaire records (housing levels 3 and 1) with different response IDs.	Q12 Q15	Pass
T14	**Database integrity constraints** Attempt a direct database insert with species Horse, age -1 and 25 hours alone.	SQLite rejects the row and the record count does not change.	A direct invalid animal insert raised SQLite IntegrityError and the animal count did not change.	Q12 Q15	Pass
T15	**Application startup** Start the finished Streamlit application and request its health endpoint.	Server reports healthy and the login page loads.	The Streamlit server started on port 8765 and the health endpoint returned 'ok'. The login page loaded with both fields and buttons.	Q6 S6	Pass
T16	**Mobile-width login page** Open the login page at a 390 x 844 mobile viewport and inspect both stacked panels.	No horizontal overflow; all login controls remain readable and usable.	At 390 x 844 pixels the panels stacked vertically. Text, fields and buttons remained readable and usable without horizontal overflow.	Q6 Q13 S6	Pass

Manual evidence still required

The following checks should be completed before the final coursework submission because they require people or environments that cannot be reproduced by an automated local test:

Check	Who or where	Evidence to collect
Adopter usability	Intended adopter	Task completion, comments and whether scores and explanations are understood.
Administrator usability	Shelter or administrator stakeholder	Add, find, edit and delete task completion plus feedback on labels and confirmation messages.
Accessibility	Keyboard-only user or structured review	Keyboard navigation, focus visibility, zoom/text scaling and contrast findings.
Final deployment	Hosted version and available browsers/devices	Screenshots, browser/device names, response time and any layout limitations.

Analysis source key

Each completed test includes the analysis evidence that made the test relevant.

Code	Evidence used
E1-E3	Competitor research: Dogs Trust, RSPCA and Petfinder.
Q1	Lifestyle factors such as space, time, cost and experience.
Q2	Animal information that must be stored and shown.
Q3	Warnings and explanations for unsuitable matches.
Q5	Administrator add, edit and delete requirements.
Q6	Ease of use.
Q8	Search and filtering.
Q9	Registration data and validation.
Q10	Compatibility weighting.
Q12	Security, user roles and protected access.
Q13	Use on the stakeholder's available devices.
Q14	Information displayed after matching.
Q15	Accurate scoring and current animal profiles.
S5	Important compatibility factors identified by survey respondents.
S6	Ease of use and clear navigation.
S7	Compatibility scores would influence decisions.
S9	Animal welfare must remain more important than a high score.